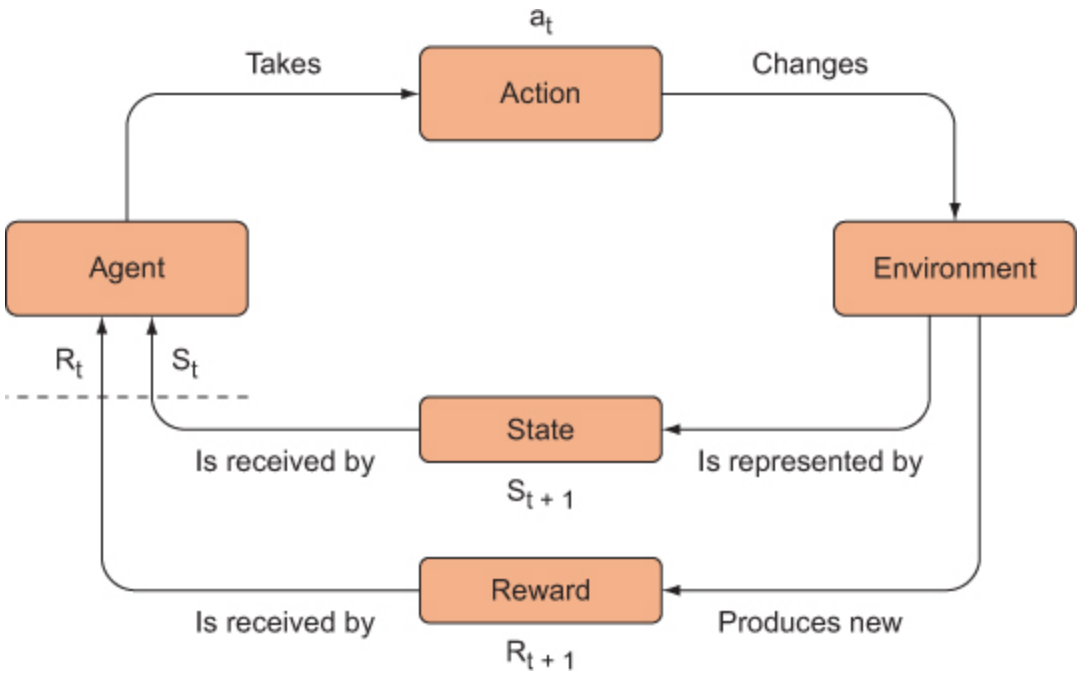
deterministically or probabilistically change the environment to be in a new state, s_{t+1}.

Figure 2.9. The general process of a reinforcement learning algorithm. The environment produces states and rewards. The agent takes an action, a_t, given a state, s_t, at time *t* and receives a reward, r_t. The agent's goal is to maximize rewards by learning to take the best actions in a given state.



The probability associated with mapping a state to a new state by taking an action is called the *transition probability*. The agent receives a reward, r_t, for having taken action a_t in state s_t leading to a new state, s_{t+1}. And we know that the ultimate goal of the agent (our reinforcement learning algorithm) is to maximize its rewards. It's really the state transition, $s_t \rightarrow s_{t+1}$, that produces the reward, not the action per se, since the action may probabilistically lead to a bad state. If you're in an action movie (no pun intended) and you jump off a roof onto another roof, you may land gracefully on the other roof or miss it completely and fall—your peril is what's important (the two possible resulting states), not the fact that you jumped (the action).

2.7.1. Policy functions